Enforcing Causality and Passivity of Neural Network Models of Broadband S-Parameters

Hakki M. Torun*, Ahmet C. Durgun†, Kemal Aygün†, and Madhavan Swaminathan*
*3D Systems Packaging Research Center (PRC), Georgia Institute of Technology, Atlanta, GA
†Assembly and Test Technology Development, Intel Corporation, Chandler, AZ

***Abstract*—This paper proposes a method to ensure that S-Parameters generated using neural network (NN) models are physically consistent and can be safely used in subsequent time-domain simulations. This is achieved by introducing causality and passivity enforcement layers as the last two layers of the NN, while minimizing their computational overhead to the training and inference of the NN model. Proposed technique is demonstrated on learning the mapping from 13 dimensional geometrical parameters of a differential plated through hole (PTH) in package core to its corresponding broadband S-Parameters up to 100 GHz.**

I. Introduction

Machine learning (ML) techniques are promising alternatives to conventional methods for performing design optimization and uncertainty quantification for signal and power integrity (SI/PI) of microelectronic packages. Most common use of ML models in SI/PI involve deriving a parametric predictive model that replaces full-wave EM solvers to predict S-Parameters for a given set of material and geometrical parameters, which are then used in time-domain simulations for further analysis.

Neural networks (NN) and Gaussian processes (GP) are commonly used models for this purpose since they have the universal approximation property. However, there is no guarantee that the predicted S-Parameters can be used in a subsequent time-domain simulation since neither of the models guarantee that the predictions preserve the physical consistency of the S-Parameters, i.e. causality and passivity. Hence, these need to be enforced at inference time after the model is trained, meaning that the training process is unaware of such requirements and possibly does not converge to a causal and passive representation. Enforcement of causality and passivity as a post-processing step then can lead to substantial computational overhead along with reduced prediction accuracy and limit their use in probabilistic analysis since the learned distribution do not represent a physical structure.

In this paper, we focus on enforcing causality and passivity of NN models during the training process with the goal of learning a physical representation between input parameters and broadband S-Parameters. We present two new layers to be used in a NN, namely *causality enforcement layer (CEL) and passivity enforcement layer (PEL)*, that ensure the predictions are responses of a causal and passive system. We demonstrate the proposed model on learning the mapping of 13 geometrical parameters of a differential plated through hole (PTH) in package core to its corresponding S-matrix up to 100 GHz.

II. Causality and Passivity Enforcement Layers

A. Causality Enforcement Layer (CEL)

S-Parameters of passive, linear time-invariant (LTI) interconnects are said to be causal if each element in the time-domain impulse response matrix can not produce an output before the input signal, i.e. $h(t) = 0$ for $t < 0$ [1]. This leads to the well-known Kramers-Kronig relations, given as

$$V(jw) = -\mathscr{H}[U(jw)] = \frac{-1}{\pi}\int_{-\infty}^{\infty}\frac{U(j\widetilde{w})}{w-\widetilde{w}}\mathrm{d}\widetilde{w} \qquad (1)$$

where $H(jw) = U(jw) + jV(jw)$ and $\mathscr{H}[\cdot]$ denotes Hilbert transform. Kramers-Kronig relations suggest that it is sufficient for a NN model to form only the real part of the S-Parameters and reconstruct the imaginary part using (1) to get a causal transfer function. However, the S-Parameters used for training is most commonly bandlimited and tabulated, which limits the direct use of (1) for such purposes.

The truncation error due to bandlimitedness can be minimized by extrapolating the data beyond the maximum available frequency. Let $U[n] = U(jw_s n)$ for $n = 0, 1, ..., N-1$ be the observed transfer function at a step size of w_s, where the maximum observed frequency is $w_{\max}= (N-1)w_s$. The principal of analytic continuation [2] states that it is possible to extrapolate $U[n]$ beyond $w_{\max}$ such that $V(jw_s n)$ can be reconstructed at observed frequencies. Here, we exploit the universal approximator property of the NN and use it to construct $U[n]$ for $N \leq n \leq MN-1$ where M is the extrapolation factor, and backpropagate through in-band reconstruction error with the goal of minimizing it.

That is, we treat the input latent variable to CEL, $X[n]$, to be the real part of an NM-point extrapolated frequency response and use an FFT based implementation of Hilbert transform [3] to construct the NM-point imaginary part, $Y[n]$, as

$$Y[n] = -\operatorname{Im}\left\{\mathscr{H}\left[X[n]\right]\right\} = -\operatorname{Im}\left\{\mathcal{F}^{-1}\left\{Z[\nu]\right\}\right\} \qquad (2)$$

where

$$Z[\nu] = \begin{cases} \widetilde{X}[0], & \nu = 0 \\ 2\widetilde{X}[\nu], & 1 \leq \nu \leq \frac{NM}{2}-1 \\ \widetilde{X}[\frac{NM}{2}], & \nu = \frac{NM}{2} \\ 0, & \frac{NM}{2} < \nu \leq NM-1 \end{cases} \qquad (3)$$

and $\widetilde{X}[\nu] = \mathcal{F}\left\{X[n]\right\}$ is the discrete Fourier transform of the input in the ν-domain. The implementation of Hilbert transform in (2) and (3) has several advantages over other techniques. First, there is no need to develop a new integration kernel to handle the singularity in (1). Another advantage is the elimination of the inaccuracies of IFFT due to bandlimited data. For instance, taking IFFT of the input sequence, enforcing $x(t) = 0$ for $t < 0$ and reconstructing back using FFT would suffer from a substantial accuracy loss due to the finite sequence length of $X[n]$. By taking the FFT of the input sequence as in (3), the constructed $\widetilde{X}[\nu]$ is forced to be periodic in ν-domain and not bandlimited, hence, an

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exact IFFT can be calculated. Note that during the training of a NN, $X[n]$ is a $N_d \times D_y \times NM$ tensor where N_d and D_y denote number of data points and output dimensionality, respectively, and all the calculations are performed efficiently in a parallelized fashion.

In order to address the discretization error, w_s can be decreased to interpolate $X[n]$ by a factor of K to an NMK-point sequence. However, such an interpolation can disrupt the causality properties. Here, the interpolation should preserve the orthogonality of $X[n]$ and $Y[n]$, i.e. $\sum_n X[n]Y[n] = 0$. The so-called causality preserving interpolation can then be achieved by padding $(NMK - NM/2 - 1)$ zeros to the end of $Z[\nu]$ in (3) and scaling the IFFT by K [3].

The CEL can then be written as a "parameterless" NN layer that takes a real tensor of size $N_d \times D_y \times NM$ as input and constructs a complex output that is the frequency response of a causal system with a frequency step size of w_s/K as

$$H[n] = f\left(X[n]\right) = K\mathcal{F}^{-1}\left\{Z[\nu]\right\} \tag{4}$$

where $H[n]$ is the $N_d \times D_y \times NMK$ sized complex output tensor of the CEL, which is forwarded to PEL.

B. Passivity Enforcement Layer (PEL)

An S-matrix is said to be passive if and only if the maximum singular value of the S-matrix is bounded at all frequencies, i.e. $\sigma_{\max}(S(jw)) \leq 1$, and Kramers-Kronig relations hold [4]. Since the input to the PEL, $H[n]$, is the complex output from the CEL, the latter is established. The former condition can be enforced using an SVD based method. However, most common SVD algorithms contain sequential bidiagonalization operations, hence, can not be easily parallelized. This would add a significant computational overhead to the training of the NN since the SVD would have to be calculated at each frequency point of each data, corresponding to $N_d NMK$ sequential SVD and gradient operations.

To limit the computational overhead, we propose to use an upper bound to $\sigma_{\max}$ that can be calculated using only matrix-matrix multiplication and Hadamard products, hence, can be massively parallelized.

Let $C(w) = tr\left(S^H(jw)S(jw)\right)$ and $D(w) = tr\left(\left(S^H(jw)S(jw)\right)^2\right)$ where $tr(\cdot)$ and $(\cdot)^H$ are the trace and Hermitian transpose operators, respectively. The $\sigma_{\max}$ of an $N \times N$ S-matrix is then bounded by [5]

$$\sigma_{\max}(w) \leq \hat{\sigma}_{\max}(w) = \sqrt{\frac{C(w)}{N} + \left(\frac{N-1}{N}\left(D(w) - \frac{C(w)^2}{N}\right)\right)^{0.5}} \tag{5}$$

To minimize the number of matrix-matrix multiplications that has greater than quadratic time complexity, $C(w)$ and $D(w)$ can be calculated efficiently using Hadamard products as

$$C(w) = \textstyle\sum_{i=1}^{N} |S_{ii}(jw)|^2 \tag{6}$$

$$D(w) = \textstyle\sum_{i,j=1}^{N} \left[\left(S^H(jw)S(jw)\right) \odot \left(S(jw)S^H(jw)\right)\right]_{ij} \tag{7}$$

where $\odot$ is the Hadamard product.

Once the $\hat{\sigma}_{\max}(w)$ is calculated for every data point in a parallelized fashion, the passivity enforcement can be performed by simply dividing the input frequency response forwarded by CEL, $H[n]$, by $\hat{\sigma}_{\max} = \max\left(\hat{\sigma}_{\max}(w)\right)$ if it is greater than 1.

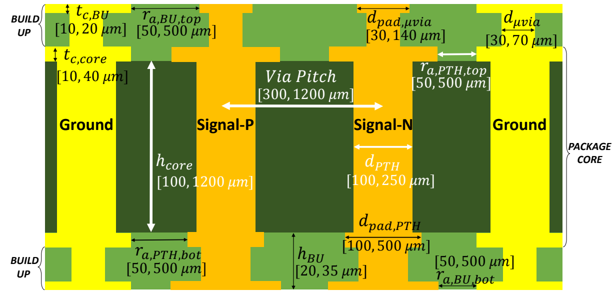

Fig. 1: Parameters of the differential PTH in package core.

Note that this enforcement can be done at individual frequency points, rather than dividing the whole frequency response, to minimize deviations between original and enforced data. However, this would disrupt the Kramers-Kronig relations and the resulting response would violate causality. As the passivity enforcement is done during the training, the proposed approach forces the training to converge to a passive and causal representation of broadband S-Parameters that minimize the error between original and predicted responses, which naturally addresses the deviation problem.

Similar to CEL, PEL can be written as a parameterless NN layer that takes $N_d \times D_y \times NMK$ complex tensor as input and constructs a complex tensor that is the frequency response of a passive and causal system as

$$\hat{S}[n] = f\left(H[n]\right) = \frac{H[n]}{\hat{\sigma}_{\max}(H[n])}, \quad n = 0, 1, ..., NK - 1 \tag{8}$$

where $\hat{S}[n]$ is the $N_d \times D_y \times NK$ sized complex output tensor that represent the predicted S-Parameters. The extrapolated part of the sequence ($NK \leq n \leq NKM - 1$) is disgarded as the training error is to be calculated at observed frequencies.

C. NN Model with PEL and CEL

The overall structure of the proposed model consists of 3 parts, namely base network, CEL and PEL. The input parameters are first passed through the base network that constructs the real part of extrapolated frequency response. This is then forwarded to CEL and PEL, respectively, to form the predicted frequency response. Although an arbitrary NN can be used as the base network, we use a recently developed model, namely Spectral Transposed Convolutional Net (S-TCNN) [6], that uses transposed convolutional layers to exploit spatial correlation in the frequency axis to predict broadband frequency responses. The weight sharing structure of S-TCNN allows to reduce the number of learnable parameters compared to fully-connected NNs and allows to extrapolate the frequency response without adding additional learnable parameters. Finally, the loss function used for the training of the overall model is the scaled ℓ^2-norm of the error between the predicted and the actual broadband S-matrix, averaged over N_d different frequency responses in the training set [6].

III. Application to Differential PTH in Package

The application chosen to evaluate the proposed method is modeling a differential PTH pair in package core along with the microvias that connect immediate build-up (BU) layers

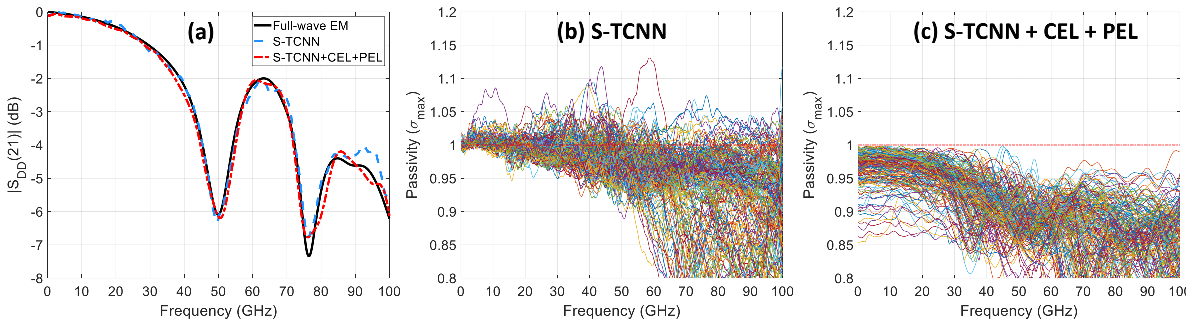

Fig. 2: Comparison of S-Parameters predicted by different models. (a) Differential insertion loss. (b, c) Passivity characterization.

to the package core. Such structures are common in off-chip high-speed channels and often times can be the bottleneck that limits the total channel bandwidth. Hence, it is important to optimize its design to minimize distortions in the signal. The objective here then is to learn a causal and passive mapping from the geometrical parameters of the PTH structure to 4-port single ended broadband S-Parameters such that it can be used later for optimization purposes. The input parameters comprise a 13 dimensional space and their corresponding bounds are given in Fig. 1. Note that the large sample space is chosen to contain various technology nodes to avoid creating different models for different manufacturing technologies.

In order to create the predictive model, 682 samples based on Latin Hypbercube Sampling (LHS) are determined. These are then fed into a full-wave EM solver to characterize their corresponding S-Parameters between 0.1-100 GHz at 100 MHz frequency steps, where the structure is excited by coaxial waveports at the antipads of microvias.

After the data is collected, 500 out of 682 samples are used for the training of the model. As the PTH structure is a partially symmetric and reciprocal system, the target data to be trained is determined to be the real and imaginary parts of the frequency responses of S_{11}, S_{12}, S_{13}, S_{14}, S_{33} and S_{34}, corresponding to a total of 12000 output dimensions. We use $K = 2$ in (2) and $M = 1.5$ in (3) to reduce the frequency step size of the predicted response to 50 MHz up to 150 GHz.

We compare the proposed methodology (S-TCNN + CEL + PEL) with S-TCNN and assess their accuracy using normalized mean squared error (NMSE) metric over each frequency response in the test set [6].

Table I summarizes the NMSE values for each model along with the causality metrics and passivity violations of the predicted S-Parameters that are calculated using a commercial tool. The predictions done by S-TCNN had an average causality quality metric of 11.17% and none of the predicted S-Parameters were causal, whereas all of the S-Parameters predicted by the proposed methodology were characterized as causal with a quality metric of 100.0%. Further, maximum singular values of the predicted S-matrices for each model are given in Fig. 2. Here, it can be seen that the PEL guarantees passivity of the predicted S-Parameters as the $\sigma_{\max}$ of all predictions are less than 0.99, while the predictions done by S-TCNN not necessarily result in a passive response as most of the predicted responses have $\sigma_{\max} > 1$.

Predicted differential insertion loss for both models are also compared to the full-wave EM solver for a random test case in Fig. 2(a). Both models showed a good prediction accuracy as the NMSE values over entire test set were 6.16% and 5.53% for S-TCNN and S-TCNN with CEL and PEL, respectively.

TABLE I: Comparison of Models on Test Data

	S-TCNN	**S-TCNN + CEL + PEL**
NMSE	6.16%	5.53%
Av. Causality Metric	11.17%	100.0%
Range of $\sigma_{\max}$	[0.653, 1.131]	[0.607, 0.999]

IV. Conclusion

In this paper, we have proposed *causality and passivity enforcement layers* to ensure physical consistency of NN models of broadband S-Parameters. In CEL, we have shown that truncation and discretization errors due to tabulated and bandlimited S-Parameter data can be minimized by utilizing analytic continuation and causality-preserving interpolation. To minimize the computational overhead of PEL to the training of the overall network, we have used an easily parallelized bound to maximum singular value of S-matrix. When applied to the problem of learning S-matrix of a PTH in package core up to 100 GHz, the proposed NN model showed 5.53% validation error without any passivity violations and had a causality metric of 100.0%, whereas the same NN without PEL and CEL had 6.16% error with an average causality metric of only 11.17% while violating passivity at multiple frequencies.

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